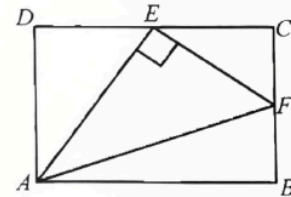


L11

Q1

- As shown in the figure below, in rectangle $ABCD$, E and F are two points on $\overline{CD}$ and $\overline{BC}$. If $m\angle AEF = 90^\circ$, then which of the following must be correct?



- A. $\triangle ADE \sim \triangle ECF$ B. $\triangle ECF \sim \triangle AEF$ C. $\triangle ADE \sim \triangle AEF$ D. $\triangle AEF \sim \triangle ABF$

Q2

- The number of correct statements among the following is _____ .

- ① all squares are similar
- ② all rhombi are similar
- ③ all rhombi with equal side lengths are similar
- ④ all rectangles with equal diagonal lengths are similar

3 According to the following sets of conditions, the number of sets which can be used to deduce that $\triangle ABC$ is similar to $\triangle A_1B_1C_1$ is _____ .

① $m\angle A = 45^\circ$, $AB = 12$, $AC = 15$, $m\angle A_1 = 45^\circ$, $A_1B_1 = 16$, $A_1C_1 = 20$;

② $AB = 12$, $BC = 15$, $AC = 24$, $A_1B_1 = 20$, $A_1C_1 = 40$, $B_1C_1 = 25$;

③ $m\angle B = m\angle B_1 = 75^\circ$, $m\angle C = 50^\circ$, $m\angle A_1 = 55^\circ$;

④ $m\angle C = m\angle C_1 = 90^\circ$, $AB = 10$, $AC = 6$, $A_1B_1 = 15$, $A_1C_1 = 9$.

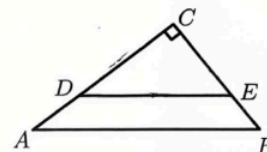
A. 1

B. 2

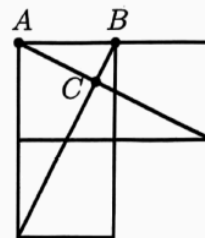
C. 3

D. 4

5 As shown in the figure, in $\triangle ABC$, $m\angle C = 90^\circ$, $AC = 4$, $BC = 3$, and $\overline{DE} \parallel \overline{AB}$. If the area of $\triangle CDE$ is equal to the area of $\triangle DABE$, calculate the lengths of $\overline{CD}$ and $\overline{DE}$.



- 7 Three unit squares and two line segments connecting two pairs of vertices are shown. What is the area of $\triangle ABC$? (2012 AMC 10A Problem, Question #15)



A. $\frac{1}{6}$

B. $\frac{1}{5}$

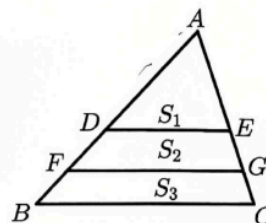
C. $\frac{2}{9}$

D. $\frac{1}{3}$

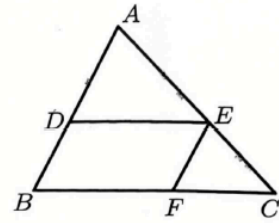
E. $\frac{\sqrt{2}}{4}$

.L11 E

- 6 As shown in the figure, $\triangle ABC$ is divided into three separate parts by $\overline{DE}$ and $\overline{FG}$, such that $S_1 = 2S_2 = 3S_3$ and that $\overline{DE} \parallel \overline{FG} \parallel \overline{BC}$. Then $DE : FG : BC = \underline{\hspace{1cm}}$.

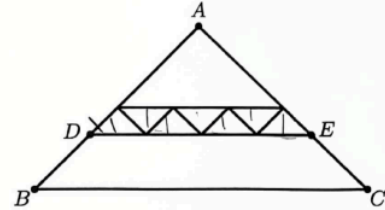


- 7 As shown in the figure, in $\triangle ABC$, point D and point E are midpoints of $\overline{AB}$ and $\overline{AC}$, respectively. Draw $\overline{DE}$, then the ratio of the area of $\triangle ADE$ to the area of quadrilateral $BCED$ is _____.



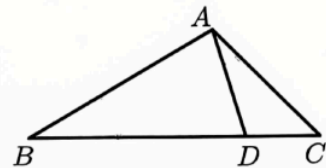
L12

- 7 All of the triangles in the diagram below are similar to isosceles triangle ABC , in which $AB = AC$. Each of the 7 smallest triangles has area 1, and $\triangle ABC$ has area 40. What is the area of trapezoid $DBCE$? (2018 AMC 10A Problem, Question #9)



- A. 16 B. 18 C. 20 D. 22 E. 24

- 5 As shown in the figure, in $\triangle ABC$, $AC = 2$, $BC = 4$, and D is a point on $\overline{BC}$. $m\angle CAD = m\angle B$. If $\triangle ADC$ has an area of a , then the area of $\triangle ABD$ is _____.

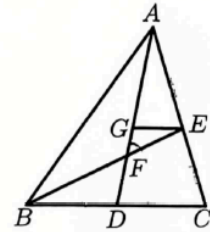


- A. $2a$ B. $\frac{5}{2}a$ C. $3a$ D. $\frac{7}{2}a$

L12 E

- 7 As shown in the figure, in $\triangle ABC$, $\overline{AD}$ and $\overline{BE}$ are medians that intersect at point F . $\overline{EG} \parallel \overline{BC}$, intersecting with $\overline{AD}$ at point G .

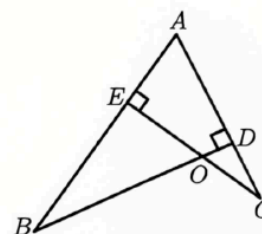
(1) Prove that $\frac{AG}{AD} = \frac{GF}{FD}$.



- 4 A square with side length 3 is inscribed in an isosceles triangle with one side of the square along the base of the triangle. A square with side length 2 has two vertices on the other square and the other two on sides of the triangle, as shown. What is the area of the triangle?(2021 AMC Fall 10B, Question #13)

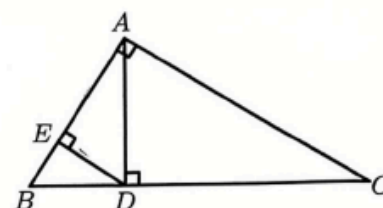
A. $19\frac{1}{4}$ B. $20\frac{1}{4}$ C. $21\frac{3}{4}$ D. $22\frac{1}{2}$ E. $23\frac{3}{4}$

- 5 As shown in the figure, $\overline{BD} \perp \overline{AC}$ at point D , and $\overline{CE} \perp \overline{AB}$ at point E . $\overline{BD}$ intersects with $\overline{CE}$ at point O . If $AB = 4$, and $AC = 3$, then which of the following statement is incorrect?



- A. $\frac{OC}{OD} = \frac{OB}{OE}$
 B. $\frac{AE}{AD} = \frac{3}{4}$
 C. $\frac{\text{Area of } \triangle ACE}{\text{Area of } \triangle ABD} = \frac{16}{9}$
 D. $\frac{\text{Perimeter of } \triangle ABD}{\text{Perimeter of } \triangle ACE} = \frac{4}{3}$

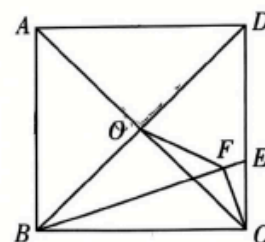
- 7 In the figure, in right triangle ABC , $m\angle BAC = 90^\circ$, $\overline{AD} \perp \overline{BC}$ at D , and $\overline{DE} \perp \overline{AB}$ at E . If $AD = 3$ and $DE = 2$, then the lengths of $\overline{AB}$ and $\overline{AC}$ are _____.



- A. $\frac{9\sqrt{5}}{5}, \frac{9}{2}$
 B. $\frac{9\sqrt{5}}{5}, \frac{21}{2}$
 C. $\frac{9}{5}, \frac{9\sqrt{5}}{2}$
 D. $\frac{9}{5}, \frac{4\sqrt{15}}{5}$

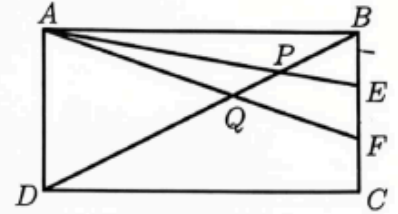
- 8 As shown in the figure, the side length of square $ABCD$ is 6. Point O is the intersection of the diagonals $\overline{AC}$ and $\overline{BD}$. Point E is on the line segment CD . Line segment CF is perpendicular to $\overline{BE}$ at point F , and the foot of the perpendicular is point F . Segment OF is then connected.

(1) Prove that $\triangle BOF \sim \triangle BED$.



(2) If $DE = 2CE$, find OF .

- 9 In rectangle $ABCD$, $AB = 6$ and $BC = 3$. Point E between B and C , and point F between E and C are such that $BE = EF = FC$. Segments $\overline{AE}$ and $\overline{AF}$ intersect $\overline{BD}$ at P and Q , respectively. The ratio $BP : PQ : QD$ can be written as $r : s : t$ where the greatest common factor of r , s and t is 1. What is $r + s + t$? (2016 AMC 10A Problem, Question #19)



A. 7

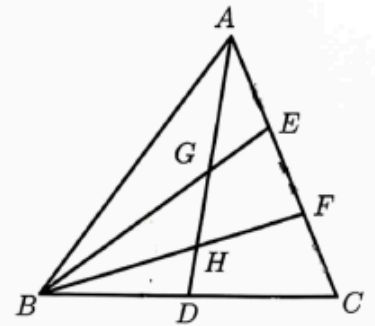
B. 9

C. 12

D. 15

E. 20

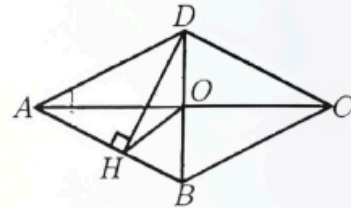
- 10 In the figure, in triangle ABC , point D is the midpoint of side $\overline{BC}$, and points E and F are the trisection points of side $\overline{AC}$. $\overline{AD}$ intersects $\overline{BE}$ and $\overline{BF}$ at points G and H , respectively. Then $AG : GH : HD = \underline{\hspace{2cm}}$.



L13

Q2

As shown in the figure, $ABCD$ is a rhombus. Its diagonals $\overline{AC}$ and $\overline{BD}$ intersect at point O , $\overline{DH} \perp \overline{AB}$ at point H , and draw $\overline{OH}$. If $m\angle CAD = 20^\circ$, then $m\angle DHO = \underline{\hspace{1cm}}$.



A. 20°

B. 25°

C. 30°

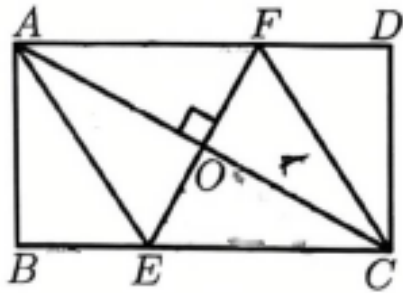
D. 40°

3 Which of the following conditions can be used to determine that a quadrilateral is a rhombus?

- A. A pair of opposite sides are parallel and equal, and one angle is a right angle.
- B. One pair of opposite sides are parallel, the other pair of opposite sides are equal, and the two diagonals are perpendicular to each other.
- C. Two pairs of opposite sides are equal respectively, and a pair of adjacent angles are equal.
- D. A pair of opposite sides are equal, a pair of adjacent sides are equal, and a pair of opposite angles are equal.

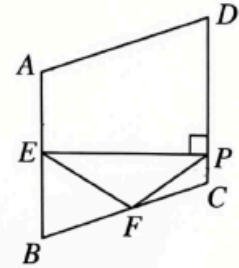
- 9 As shown in the figure, in the rectangle $ABCD$, O is the midpoint of the diagonal AC . A line is drawn through point O , intersecting the sides BC and AD at points E and F , respectively. Connect AE and CF .

(2) If $AB = 6$, $BC = 8$, find the side length of rhombus $AECF$.



L13 E

- 1 As shown in the figure, in rhombus $ABCD$, $m\angle A = 110^\circ$, points E and F are midpoints of $\overline{AB}$ and $\overline{BC}$, respectively. $\overline{EP} \perp \overline{CD}$ at point P . Then $m\angle PEF = \underline{\hspace{1cm}}$.



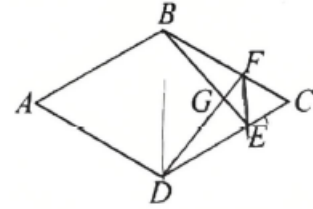
A. 35°

B. 45°

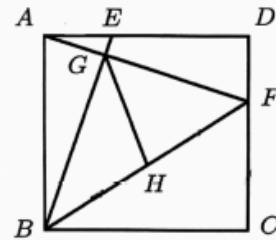
C. 50°

D. 55°

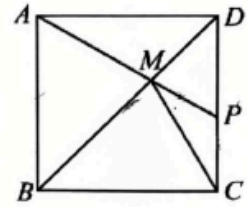
- 5) As shown in the diagram, in the rhombus $ABCD$, E , F are points on sides CD and BC , respectively. $CE = CF$. Connect BE and DF , intersecting at point G . Prove that $BE = DF$.



- 7) The side length of the square $ABCD$ is 8, points E and F are on AD and CD respectively, $AE = DF = 2$, BE intersects with AF at point G , point H is the midpoint of BF . Connect GH , then the length of GH is _____.

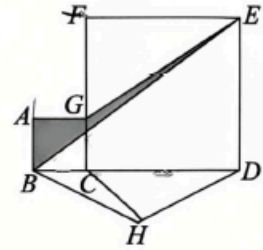


- 8 As shown in the figure, in the square $ABCD$ with side length 6, M is a point on the diagonal BD . Connect AM and extend it to intersect CD at point P . If $PM = PC$, what is the length of AM ?

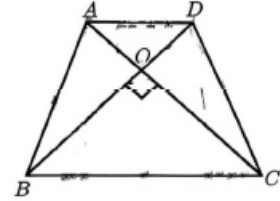


- A. $3(\sqrt{3} - 1)$ B. $3(3\sqrt{3} - 2)$ C. $6(\sqrt{3} - 1)$ D. $6(3\sqrt{3} - 2)$

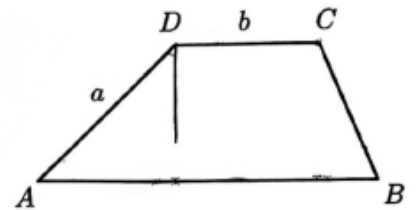
- 9 As shown in the figure, in square $ABCG$ and square $CDEF$, point G is the trisection point of CF , point H and point A are centrosymmetric to point C . Connect EG , BE , BH , DH . If the shaded area is 4, then the length of BH is _____ .



- 2 As shown in the figure, in isosceles trapezoid $ABCD$, $\overline{AD} \parallel \overline{BC}$. $AB = DC$, $\overline{AC} \perp \overline{BD}$. If $AD = 4$ and $BC = 6$, then the area of $ABCD$ is _____.

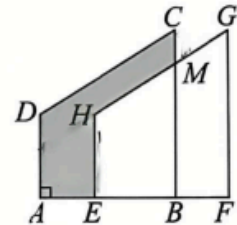


- 3 In quadrilateral $ABCD$, it is given that $\overline{AB} \parallel \overline{CD}$ and that $m\angle D = 2m\angle B$. Lengths of $\overline{AD}$ and $\overline{CD}$ are equal to a and b , respectively. Then $AB =$ _____.



- 6 In the right trapezoid $ABCD$, $AD \parallel BC$, $\angle A = 90^\circ$, $AB = 3$, $CD = 6$, then the measure of $\angle D$ is _____ .

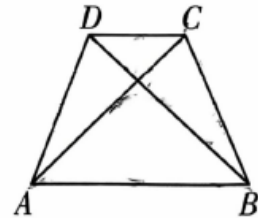
- 7 As shown in the figure, in the right trapezoid $ABCD$, $AD \parallel BC$, $AD \perp AB$, $BC = 5$, the right trapezoid $ABCD$ is translated 2 units along the AB direction to obtain the right trapezoid $EFGH$, HG intersects BC at point M , and $CM = 1$. Then, the area of the shaded part in the figure is _____ .



L14 E

- 2 It is given that the ratio of the lengths of the upper and lower bases of a trapezoid is $1 : 2$. The median divides the trapezoid into two parts. The ratio of the area of the upper part to that of the lower part is _____ .

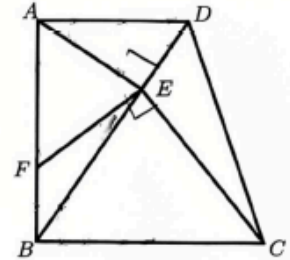
- 3 As shown in the figure, in trapezoid $ABCD$, $\overline{AB} \parallel \overline{CD}$, $AD = BC$, diagonals $\overline{AC} \perp \overline{BD}$, and $AC = 5\sqrt{2}$. Then the median of $ABCD$ has a length of _____ .



- 6 In the right trapezoid $ABCD$, $AD \parallel BC$, $AB \perp BC$, E is the midpoint of side CD , and satisfies $AB = AD + BC$, $BE = 3$. What is the area of trapezoid $ABCD$?

- 7 As shown in the figure, in the right trapezoid $ABCD$, $AD \parallel BC$, $\angle ABC = 90^\circ$, $AE \perp BD$, the foot of the perpendicular is E . Connect CE , and draw $EF \perp CE$, intersecting side AB at point F .

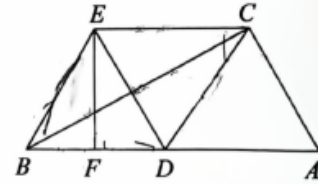
(1) Prove that: $\triangle AEF \sim \triangle BEC$.



(2) If $AB = BC$, prove that: $AF = AD$.

- 8 As shown in the figure, in $\text{Rt}\triangle ABC$, $m\angle ACB = 90^\circ$. Point D is on $\overline{AB}$. Draw $\overline{CD}$, $\overline{CE} \parallel \overline{AB}$, and $\overline{BE} \parallel \overline{CD}$, such that $CE = AD$.

(1) Prove that quadrilateral $BDCE$ is a rhombus.



- (2) Draw $\overline{EF} \perp \overline{BD}$ through point E such that the perpendicular foot is located at F . If F is the midpoint of $\overline{BD}$, and $EB = 6$, calculate the length of $\overline{BC}$.